Enabling Team Collaboration

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Module Overview

Explore state data contents

- terraform state command

State backends

Migrating state data

Globomantics Environment

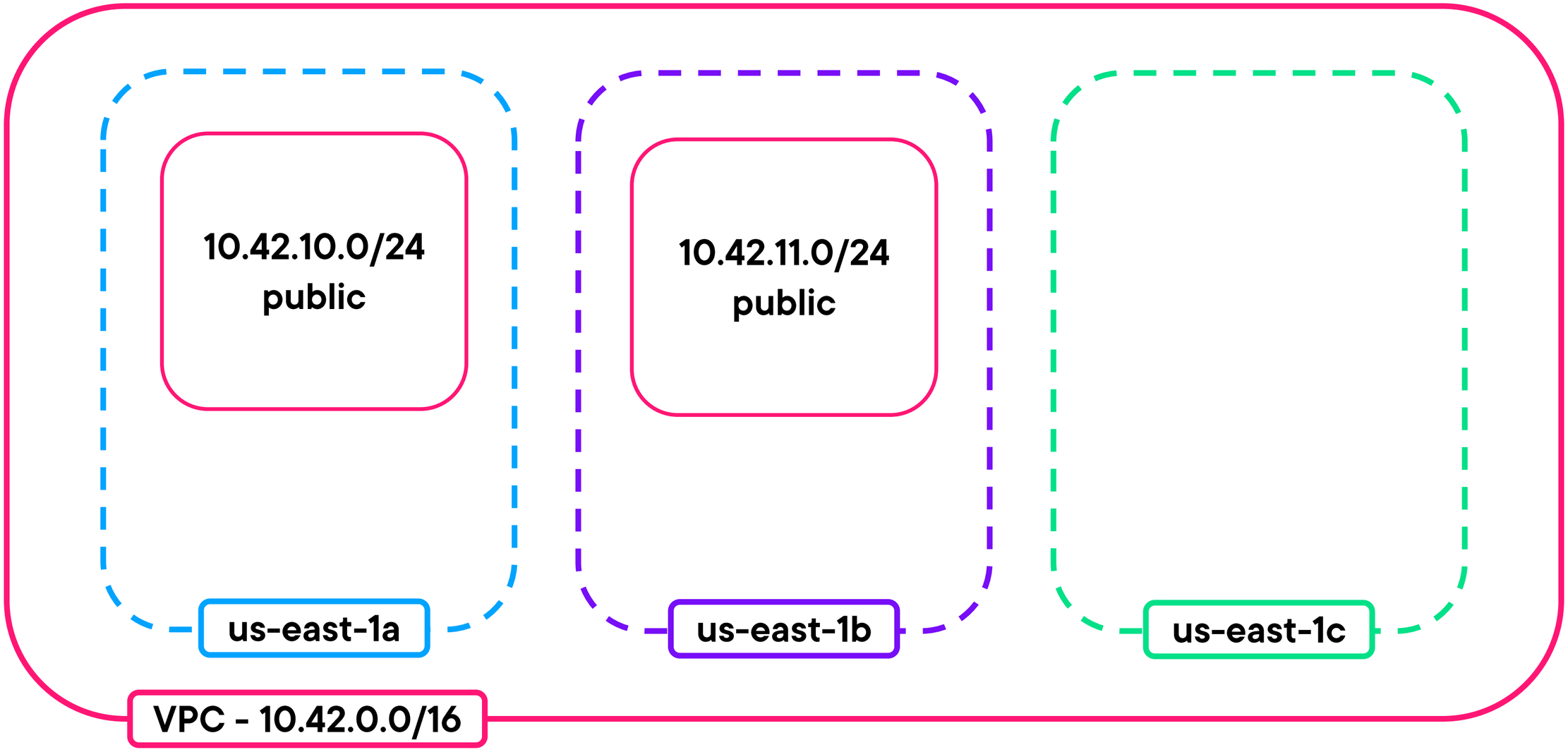
Collaborate with networking team

Share information with application teams

Enable state data access

Leverage Terraform Cloud for remote state

Existing Environment



Local State Data

terraform.tfstate

```
{
  "resources": [
    {
      "mode": "managed",
      "type": "aws_vpc",
      "name": "main",
      "instances": [
        "id": "vpc-123654789"
      ]
    }
  ]
}
```

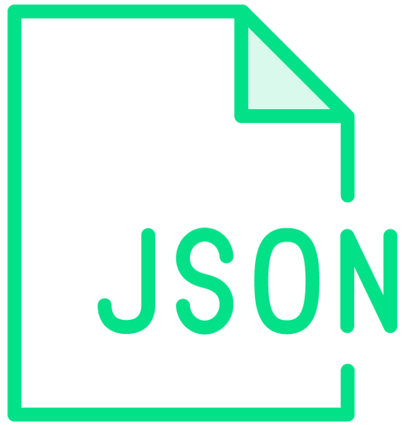
main.tf

```
module "main" {
  source      = "terraform-aws-..."
  version     = "5.0.0"

  cidr_block = var.vpc_cidr

}
```

Terraform State Data



Mapping of configuration to actual resources

Stored in JSON (don't touch!)

State contents

- Metadata
- Resources, data sources, and outputs

```
{
  ...
  "resources": [
    {
      "mode": "data",
      "type": "aws_availability_zones",
      "name": "available",
      "instances": [{
        "id": "us-east-1",
        ...
      }]
    }
  ]
}
```

- Resource list
- Mode set to data for a data source

```
{
  ...
  "resources": [
    {
      "module": "module.main",
      "mode": "managed",
      "type": "aws_vpc",
      "name": "this",
      "instances": [{
        "id": "vpc-123654789",
        ...
      }]
    }
  ]
}
```

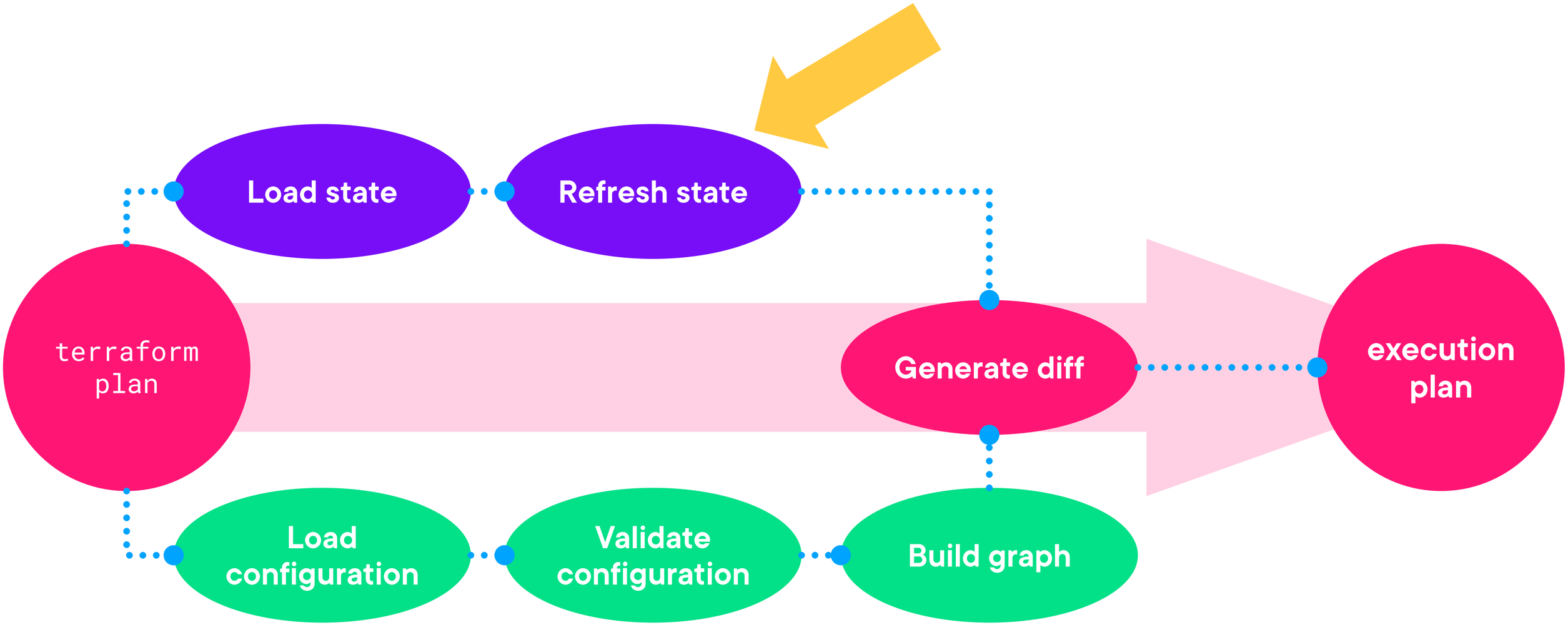
- Resource list
- Module address
- Mode set to managed for a resource
- Type set to "aws_vpc"
- Name set to "this"
- Instances at the resource address
- ID set to actual resource

```
{
  ...
  "outputs": {
    "vpc_id": {
      "value": "vpc-123654789",
      "type": "string"
    }
  }
}
```

- Outputs map
- Output name
- Value of output
- Type of output

Interacting with State Data

Terraform Planning Process



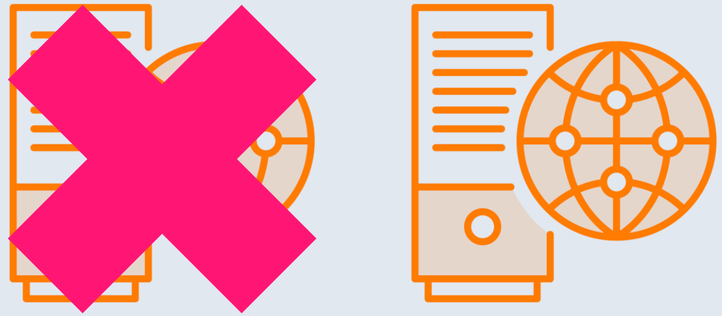
Refreshing State Data

```
# Refresh state data based on managed resources
terraform refresh

# Refresh state data only as part of execution plan
terraform plan -refresh-only

terraform apply -refresh-only
```

Recreating Resources



```
# Mark a resources for recreation in state
terraform taint ADDR
terraform taint aws_instance.web

# Replace a resource as part of the execution plan
terraform plan -replace="aws_instance.web"

terraform apply -replace="aws_instance.web"
```

Moving Resources

```
terraform state mv aws_vpc.main aws_vpc.web
```

main.tf

```
resource "aws_vpc" "main" {
  cidr_block           = var.vpc_cidr
  enable_dns_support = true

  tags = {
    Name = var.vpc_name
  }
}
```

main.tf

```
resource "aws_vpc" "web" {
  cidr_block           = var.vpc_cidr
  enable_dns_support = true

  tags = {
    Name = var.vpc_name
  }
}
```

Moving Resources

```
# Move resources from old to new address
terraform state mv OLD_ADDR NEW_ADDR
terraform state mv aws_vpc.main aws_vpc.web

# Moved block
moved {
  from = OLD_ADDR
  to   = NEW_ADDR
}

moved {
  from = aws_vpc.main
  to   = aws_vpc.web
}
```

Removing Resources

```
# Remove the resource entry at the given address
terraform state rm ADDR
terraform state rm aws_vpc.main

# Also must remove from configuration
```

Other State Commands

```
# List all resources and data sources
terraform state list

# List all attributes of a single object
terraform state show ADDR
terraform state show aws_vpc.main

# Send a copy of state data to stdout
terraform state pull

# Push state data to a path
terraform state push PATH
```

Terraform State Backends

Backend Options

Local backend

- Default setting

Remote backend

- S3, Azure Storage, Google Cloud Storage
- Terraform Cloud and Consul

Features

- Locking
- Workspaces

Data encryption

Access control

Backend Block

backend.tf

```
terraform {
  backend "backend_type" {
    IDENTIFIER = VALUE
  }
}
```

Backend Block

backend.tf

```
terraform {

  backend "s3" {

    bucket         = "globo-state-12345"
    key            = "dev/network/terraform.tfstate"
    dynamodb_table = "globo-state-12345"
    region         = "us-east-1"
    access_key     = "HGT657UHT56786YUHT6"
    secret_key     = "ytgvbJU&*ijhgt6%TYUijhy"

  }

}
```

Backend Block

backend.tf

```
terraform {

  backend "s3" {

    bucket         = "globo-state-12345"
    key            = "dev/network/terraform.tfstate"
    dynamodb_table = "globo-state-12345"

  }

}
```

```
terraform {

  backend "s3" {}

}
```

Partial Configuration

```
# Using in-line settings
terraform init -backend-config="bucket=globo-state-12345"

# Using a backend config file
terraform init -backend-config="backend-settings.txt"
```

Backend Planning

Globomantics Environment

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Leverage Terraform Cloud for remote state

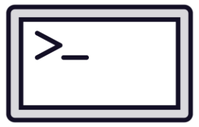
Cloud Block

backend.tf

```
terraform {

  cloud {

    organization = "globo-deep-dive"

    workspaces {
      name = "web-network-dev"
    }
  }

}
```

Migrating State Data

Update configuration with backend or cloud block



Run terraform init with any required flags

Confirm state data migration

Module Summary

Terraform state is kinda important

Declarative state management

Remote state for security and collaboration

Up Next:

Building a CI/CD Workflow